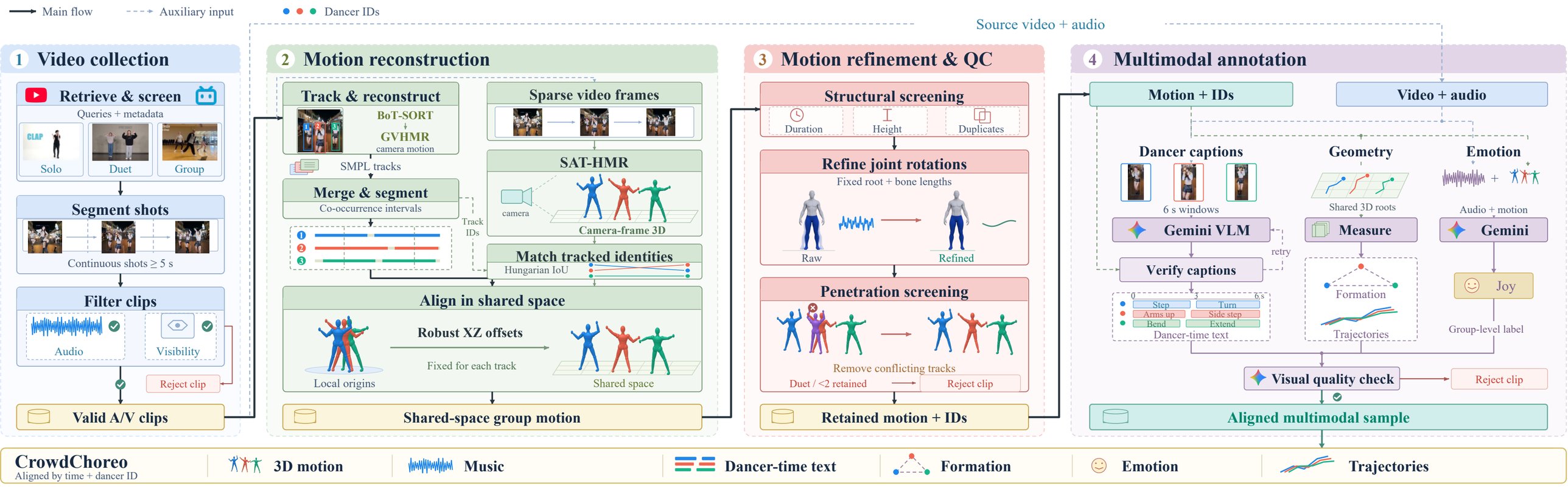

Main flow
Auxiliary input
Dancer IDs
Source video + audio
1 Video collection
Retrieve & screen
Queries + metadata
CLAP
Solo
Duet
Group
Segment shots
Continuous shots ≥ 5 s
Filter clips
Audio
Visibility
Reject clip
Valid A/V clips
2 Motion reconstruction
Track & reconstruct
BoT-SORT
GVHMR
camera motion
SMPL tracks
Merge & segment
Co-occurrence intervals
Track IDs
Sparse video frames
SAT-HMR
camera
Camera-frame 3D
Match tracked identities
Hungarian IoU
Align in shared space
Robust XZ offsets
Fixed for each track
Local origins
Shared space
Shared-space group motion
3 Motion refinement & QC
Structural screening
Duration
Height
Duplicates
Refine joint rotations
Fixed root + bone lengths
Raw
Refined
Penetration screening
Remove conflicting tracks
Duet / <2 retained
Reject clip
Retained motion + IDs
4 Multimodal annotation
Motion + IDs
Video + audio
Dancer captions
6 s windows
Gemini VLM
retry
Verify captions
0
3
6 s
Step
Turn
Arms up
Side step
Bend
Extend
Dancer-time text
Geometry
Shared 3D roots
Measure
Formation
Trajectories
Emotion
Audio + motion
Gemini
Joy
Group-level label
Visual quality check
Reject clip
Aligned multimodal sample
CrowdChoreo
Aligned by time + dancer ID
3D motion
Music
Dancer-time text
Formation
Emotion
Trajectories